Supply an island lighthouse for a full night

Task 4.7 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

The island lamp draws 1,000 W. Assume 12 h of darkness first. Compare two of these options: delivered batteries, solar with storage, wind with storage or a fuel generator. Consider service visits and a longer night.

Compute daily demand as lamp power × operating hours. Example: 10 W for 12 h uses 120 Wh. Keep W and Wh distinct.

Case	Lamp power (W)	Hours	Demand (Wh and kWh)
12-hour night			
Longer night			

02 Plan it on paper

Compare two candidate supplies. Include operating hours, storage, site conditions, and an assumption.

My prediction, chosen feature or design limit

Supply an island lighthouse for a full night

Task 4.7 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Prepare a proposal with a labeled energy-flow diagram, demand estimate, and a plan for hours when the source is unavailable. Have a partner change one condition such as a longer night. Recalculate and revise the recommendation.

Candidate	Supply and storage	Service or site limit	How it covers darkness
Option A			
Option B			

04 Use the reading

Ch. 31 | Sunny Side Up | printed pp. 281-287

Ch. 32 | Cape Wind | printed pp. 289-294

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

What source and storage plan would you recommend? Cite one calculated demand and one site constraint.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.